Begum Rokeya Univesity,Rangpur

Department of Statistics

STAT-2205: R Programming Project

Flight Data Analysis

ID: 12310066

#setting home directory

```
getwd()

list.files()

setwd("C:/Users/USER/Downloads/R & Python")
```

##problem with source

```
.libPaths()

.libPaths("C:/Users/USER/Downloads/R & Python/pakage")


install.packages("ggplot2")

library(ggplot2)
```

 # Load dataset

```
flight <- read.csv("C:/Users/USER/Downloads/R & Python/flight_data.csv")
```

 # Random sample using student ID

```
set.seed(66)

flight_sample <- flight[sample(nrow(flight), 10000), ]
```

##Question 1 :What do coach ticket prices look like? What are the high and low values? What would be consideredthe average? Does $500 seem like a good price for a coach ticket?”

Solution:

 # Summary statistics

```
summary(flight_sample$coach_price)
```

Output

```
Min. 1st Qu.  Median    Mean 3rd Qu. 
  84.5   331.3   380.3   376.0   426.4 
  Max. 
 586.0 
```

 # Mean

```
mean(flight_sample$coach_price)
```

Output

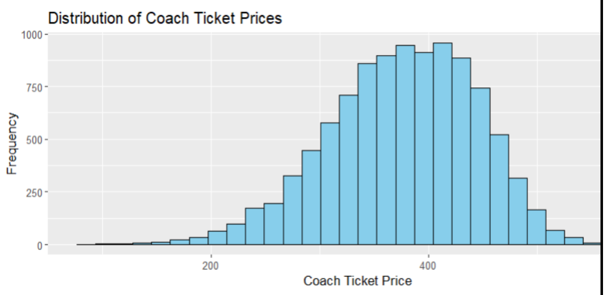
```
[1] 375.989
```

 # Histogram

```
ggplot(flight_sample, aes(x=coach_price)) +

  geom_histogram(bins=30, fill="skyblue", color="black") +

  labs(
```

```
    title="Distribution of Coach Ticket Prices",
    x="Coach Ticket Price",
    y="Frequency"
  )
```

Output



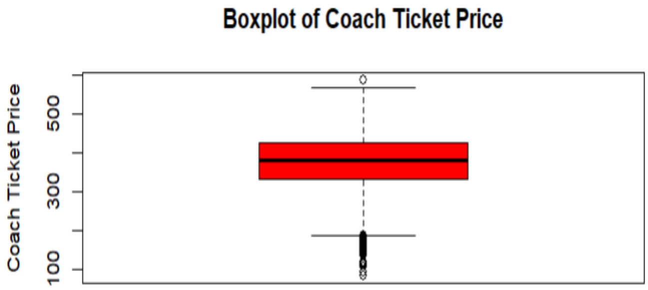
Interpretation

The histogram and summary statistics show the overall distribution of coach ticket prices. The average price can be compared with $500 to determine whether it is considered expensive or reasonable.

```
#Boxplot
boxplot(flight_sample$coach_price,
        main = "Boxplot of Coach Ticket Price",
        ylab = "Coach Ticket Price",
        col = "red",
        border = "black")
```

Output

library(ggplot2) library(ggplot2)



##Q2: What are the coach prices for flights that are 8 hours long? Does a $500 ticket seem more reasonable for these flights?

Solution:

```
#8-hour flights
flight_8 <- subset(flight_sample, hours == 8)


# Summary statistics
summary(flight_8$coach_price)
```

Output

```
  Min. 1st Qu.  Median    Mean 3rd Qu.
 213.0   383.1   426.2   426.8   477.4
  Max.
 568.5
```

Interpretation :

Long-duration flights generally have higher ticket prices. Therefore, a $500 coach ticket may be more acceptable for 8-hour flights.

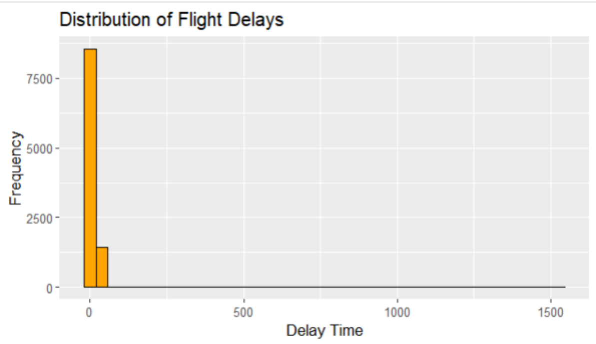
##Q3: What does the distribution of flight delays look like? Are there any outliers?

Solution:

```
library(ggplot2)
```

```
# Histogram
ggplot(flight_sample, aes(x=delay)) +
  geom_histogram(bins=40, fill="orange", color="black") +
  labs(
    title="Distribution of Flight Delays",
    x="Delay Time",
    y="Frequency"
  )
```

Output




```
# Summary
summary(flight_sample$delay)
```

Output

```
Min. 1st Qu.  Median    Mean 3rd Qu.
   0.00    9.00   10.00   13.29   13.00
   Max.
1528.00
```

Interpretation :

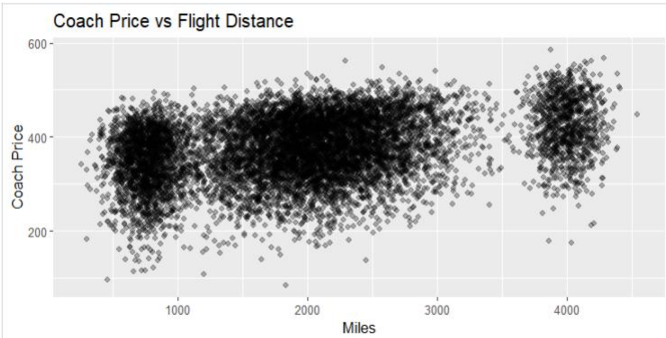
Thedelaydistributionmaycontainoutliersduetoextremedelaysexperiencedbysome flights.

##Q4: How are coach ticket prices related to flight distance (miles)?

Solution:

```
library(ggplot2)
# Scatter plot
ggplot(flight_sample,
       aes(x=miles, y=coach_price)) +
  geom_point(alpha=0.3) +
  labs(
    title="Coach Price vs Flight Distance",
    x="Miles",
    y="Coach Price"
  )
```

Output




```
# Correlation
cor(flight_sample$miles,
    flight_sample$coach_price)
```

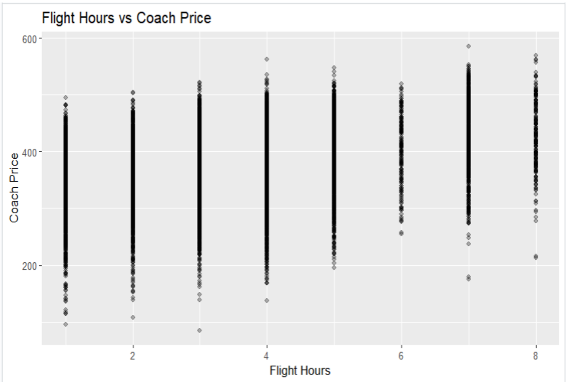
Interpretation
Thereisusuallyapositiverelationshipbetweenflightdistanceandticketprice.

##Q5: How are flight hours related to coach ticket prices?

Solution:

```r
library(ggplot2)

# Scatter plot

ggplot(flight_sample,

       aes(x=hours, y=coach_price)) +

  geom_point(alpha=0.3) +

  labs(

    title="Flight Hours vs Coach Price",

    x="Flight Hours",

    y="Coach Price"

  )
```

Output




```r
# Correlation

cor(flight_sample$hours,

    flight_sample$coach_price)
```

Output

[1] 0.3371431

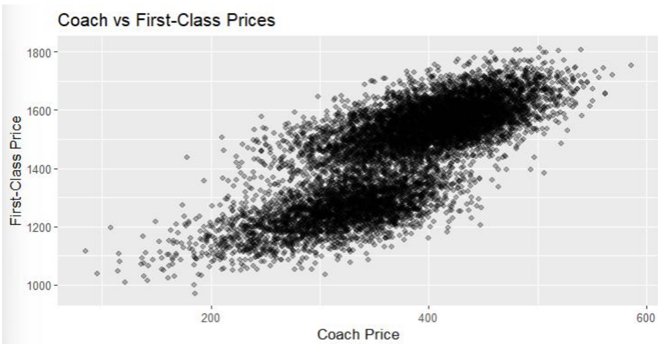
Interpretation
Longerflightdurationtendstoincreasecoachticketprices.

##Q6: How are coach ticket prices related to first-class ticket prices?

Solution:

```r
library(ggplot2)

# Scatter plot

ggplot(flight_sample,

       aes(x=coach_price,

           y=firstclass_price)) +

  geom_point(alpha=0.3) +

  labs(

    title="Coach vs First-Class Prices",

    x="Coach Price",

    y="First-Class Price"

  )
```

Output




```r
# Correlation

cor(flight_sample$coach_price,

    flight_sample$firstclass_price)
```

Output

<[1] 0.7625297 (strong possitive) >

Interpretation
Highercoachpricesaregenerallyassociatedwith higherfirst-classprices

##Q7: How do in-flight features affect coach ticket prices?

Solution:

 # Meal

```
aggregate(coach_price ~ inflight_meal,
          data=flight_sample,
          mean)
```

Output

```
 <inflight_meal coach_price
1          No    369.4530
2         Yes    390.9813>
```

 # Entertainment

```
aggregate(coach_price ~ inflight_entertainment,
          data=flight_sample,
          mean)
```

Output

```
 inflight_entertainment coach_price
1                      No     319.6572
2                     Yes     389.5651
```

 # Wifi

```
aggregate(coach_price ~ inflight_wifi,
          data=flight_sample,
          mean)
```

Output

```
inflight_wifi coach_price
1            No     311.4042
2           Yes     383.3489
```

Interpretation
Flightswithadditionalservicessuchasmeals,entertainment,andWiFioftenhavehigher ticketprices

##Q8: How do weekend flights compare to weekday flights in terms of ticket prices?

Solution:

```
library(ggplot2)
install.packages("dplyr")
library(dplyr)
# Average prices
weekend_price <- flight_sample %>%
  group_by(weekend) %>%
  summarise(
    avg_coach = mean(coach_price),
    avg_firstclass = mean(firstclass_price)
  )
print(weekend_price)
```

Output

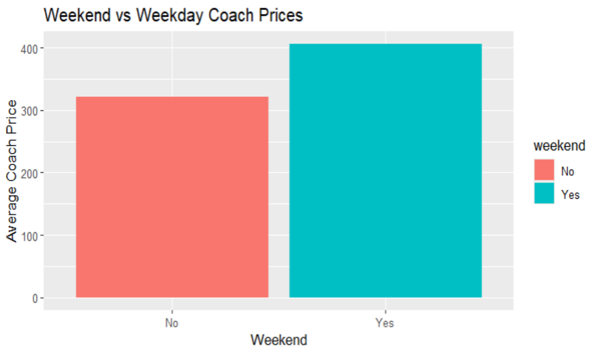
```
# A tibble: 2 × 3
  weekend avg_coach avg_firstclass
  <chr>      <dbl>          <dbl>
1 No          321.          1259.
2 Yes         406.          1559.
```

 # Bar plot

```
ggplot(weekend_price,
       aes(x=weekend,
           y=avg_coach,
           fill=weekend)) +
```

```
  geom_bar(stat="identity") +
  labs(
    title="Weekend vs Weekday Coach Prices",
    x="Weekend",
    y="Average Coach Price"
  )
```

Output



Interpretation

Weekend ticket prices may differ from weekday prices depending on passenger demand

##Q9: How do redeye flights compare to non-redeye flights in terms of coach ticket prices?

Solution :

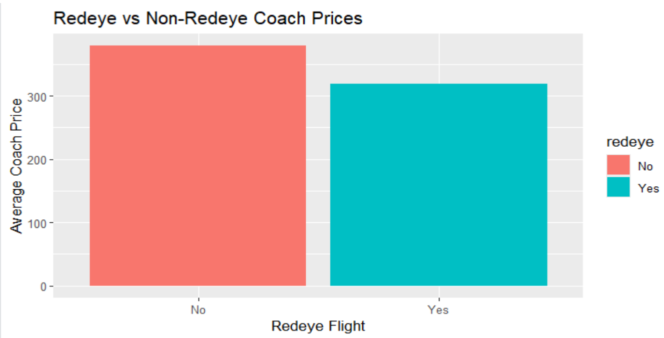
```
library(ggplot2)
library(dplyr)
# Average prices
redeye_price <- flight_sample %>%
  group_by(redeye) %>%
  summarise(
    avg_coach = mean(coach_price)
  )
print(redeye_price)
```

Output

```
# A tibble: 2 × 2
  redeye avg_coach
  <chr>      <dbl>
1 No         379.
2 Yes        318.
```

```
# Bar plot
ggplot(redeye_price,
       aes(x=redeye,
           y=avg_coach,
           fill=redeye)) +
  geom_bar(stat="identity") +
  labs(
    title="Redeye vs Non-Redeye Coach Prices",
    x="Redeye Flight",
    y="Average Coach Price"
  )
```

Output



Interpretation

Redeye flights are often cheaper because they operate during late-night hours.

##Q10: Perform the analysis for appropriate variables

Solution :

#Summary statistics

summary(flight_sample)

Output

```
          miles       passengers       delay
 Min.        : 260   Min.   :141.0   Min.   :    0.00
 1st Qu.    :1340   1st Qu.:204.0   1st Qu.:    9.00
 Median    :1978   Median :210.0   Median :   10.00
 Mean       :2001   Mean   :207.6   Mean   :   13.29
 3rd Qu.    :2468   3rd Qu.:215.0   3rd Qu.:   13.00
 Max.       :4545   Max.   :244.0   Max.   :1528.00
 inflight_meal      inflight_entertainment inflight_wifi
 Length:10000       Length:10000           Length:10000
 Class :character   Class :character       Class :character
Mode  :character   Mode  :character        Mode  :charact
er



 day_of_week         redeye            weekend
 Length:10000       Length:10000       Length:10000
 Class :character   Class :character   Class :character
 Mode  :character   Mode  :character   Mode  :character



   coach_price    firstclass_price     hours
 Min.   : 84.5   Min.   : 970.7   Min.   :1.000
 1st Qu.:331.3   1st Qu.:1300.6   1st Qu.:2.000
 Median :380.3   Median :1501.2   Median :4.000
 Mean   :376.0   Mean   :1453.7   Mean   :3.622
 3rd Qu.:426.4   3rd Qu.:1581.8   3rd Qu.:4.000
 Max.   :586.0   Max.   :1812.7   Max.   :8.000
```

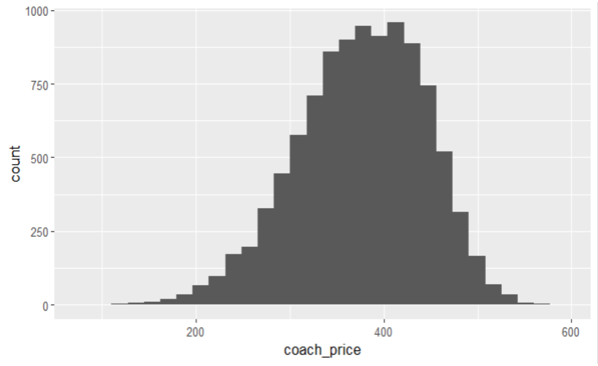
#(b) Visualization

library(ggplot2)

Histogram

```
ggplot(flight_sample,
       aes(x=coach_price)) +
  geom_histogram(bins=30)
```

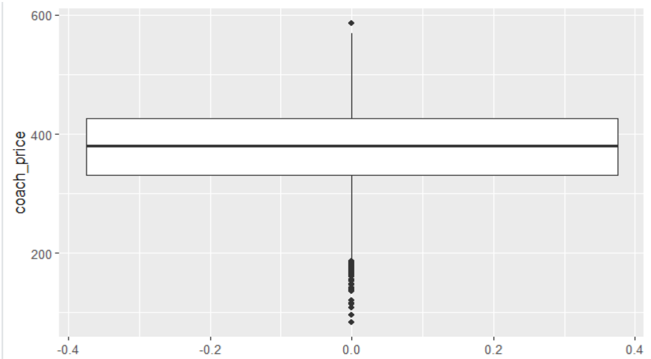
Output



Boxplot

```
ggplot(flight_sample,
       aes(y=coach_price)) +
  geom_boxplot()
```

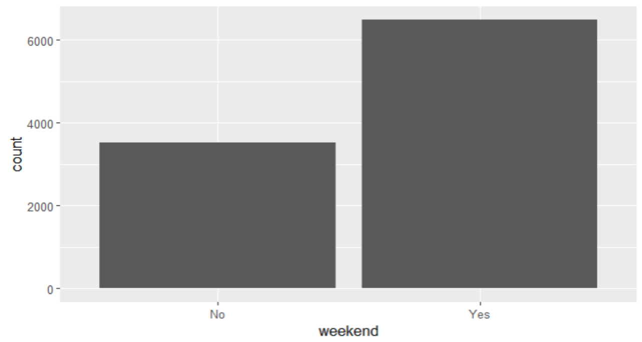
Output



Bar chart

```
ggplot(flight_sample,
       aes(x=weekend)) +
  geom_bar()
```

Output



##(c) Hypothesis Testing (Independent t-test)

```
t.test(
  coach_price ~ weekend,
  data=flight_sample
)
```

Output

```
        Welch Two Sample t-test

data:  coach_price by weekend
t = -73.684, df = 7064.8, p-value <

2.2e-16
alternative hypothesis: true difference in means betwee
n group No and group Yes is not equal to 0
95 percent confidence interval:
 -87.21771 -82.69724
sample estimates:
 mean in group No mean in group Yes
        320.8771          405.8346
```

##(d) Correlation Analysis

```
cor(
  flight_sample[, c(
    "coach_price",
    "miles",
    "hours",
    "delay",
    "firstclass_price"
  )]
)
```

Output

```
                 coach_price        miles
coach_price        1.0000000  0.342708985
miles              0.3427090  1.000000000
hours              0.3371431  0.985211774
delay             -0.0153009 -0.005755326
firstclass_price   0.7625297  0.295900806
                       hours        delay
coach_price       0.33714307 -0.015300901
miles             0.98521177 -0.005755326
hours             1.00000000 -0.004757490
delay            -0.00475749  1.000000000
firstclass_price  0.29146142 -0.025528080
                 firstclass_price
coach_price            0.76252966
miles                  0.29590081
hours                  0.29146142
delay                 -0.02552808
firstclass_price       1.00000000
```

##(e) Linear Regression

```
model <- lm(
  coach_price ~ miles + hours + delay,
  data=flight_sample
)
summary(model)
```

Output

```
Call:
lm(formula = coach_price ~ miles + hours + d
elay, data = flight_sample)

Residuals:
     Min       1Q   Median       3Q      Max
-287.736  -42.123    4.627   49.941  178.604

Coefficients:
              Estimate Std. Error t value
(Intercept) 326.852093   1.512646 216.080
miles         0.025869   0.003952   6.546
hours        -0.647266   2.145323  -0.302
delay        -0.021212   0.014969  -1.417
            Pr(>|t|)
(Intercept)  < 2e-16 ***
miles       6.21e-11 ***
hours          0.763
delay          0.157
---
Signif. codes:
0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

```
Residual standard error: 63.95 on 9996 degrees of freedom
Multiple R-squared:  0.1176,	Adjusted R-squared:  0.1174
F-statistic: 444.2 on 3 and 9996 DF,  p-value: < 2.2e-16
```

#10(f) Logistic Regression

```
flight_sample$redeye_binary <-
  ifelse(
    flight_sample$redeye=="Yes",
    1,0
  )

log_model <- glm(
  redeye_binary ~ hours + coach_price,
  data=flight_sample,
  family="binomial"
)
summary(log_model)
```

Output

```
Call:
glm(formula = redeye_binary ~ hours + coach_price, family = "binomial",
    data = flight_sample)

Coefficients:
              Estimate Std. Error z value
(Intercept)  1.2025056  0.2130049   5.645
hours        0.2716457  0.0276520   9.824
coach_price -0.0144507  0.0006844 -21.114
            Pr(>|z|)
(Intercept) 1.65e-08 ***
hours        < 2e-16 ***
coach_price  < 2e-16 ***
---
Signif. codes:
0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 4254.0  on 9999  degrees of freedom
Residual deviance: 3767.7  on 9997  degrees of freedom
AIC: 3773.7

Number of Fisher Scoring iterations: 6
```

Conclusion

This analysis explored several important aspects of airline ticket pricing and flight charac teristics using R programming. The study identified relationships between ticket prices, distance, flight duration, delays, and additional flight services. Statistical analysis and visualization techniques helped provide meaningful insights into airline pricing patterns.

References

- RProgramming Documentation
- ggplot2 Package Documentation
- dplyr Package Documentation